Altered calcium activity patterns in developing dopaminergic neurons from PD patients as an early indicator of disease predisposition or progression

Project 136

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Summary

The main goal of this project was the identification of potentially aberrant expression patterns and activities of 66 calcium (Ca^{2+})-associated candidate genes [1] in developing midbrain dopaminergic (mDA) neurons derived from human induced pluripotent stem cells (hiPSCs) of idiopathic PD (iPD) patients vs. healthy controls (HC). We hypothesized that aberrant gene expression or Ca^{2+}-activity patterns detected for some of these 66 Ca^{2+}-associated candidate genes in developing mDA neurons derived from iPD hiPSCs compared to HC hiPSCs might serve as an early indicator of disease predisposition or progression. To this end, we established the directed and reproducible differentiation of these iPD and HC hiPSCs into mDA neurons in the culture dish [2]. We also used a stringent quality control based on the monitoring of 31 marker genes in these cells throughout the differentiation procedure. Subsequently or in parallel, the expression of 64 out of the 66 candidate genes (due to the inability to design proper primers for two of these genes) was monitored in the developing mDA cells by quantitative reverse transcription and polymerase chain reaction (RT-qPCR) of total RNA harvested at three different time-points of the differentiation procedure: from hiPSCs (d0), mDA precursors (d30) and mature mDA neurons (d60).

Method

1. *hiPSC Acquisition and Cell Culture*

Nine (9) hiPSC lines (passage number 10 to 11) were obtained from the PPMI cohort through a MJFF Access to Data & Biospecimens 2017 award to NP. Of these, 5 PPMI hiPSC lines were from patients with idiopathic (sporadic) PD (iPD) and 4 PPMI hiPSC lines were from healthy controls (HC). Recipients were blinded to the identity of the PPMI hiPSC lines throughout the analyses.

PPMI hiPSCs were grown in StemMACS™ iPS-Brew medium (#130-104-368; Miltenyi Biotec, Bergisch Gladbach, Germany) on hESC-qualified Matrigel (#354277; Corning, Amsterdam, The Netherlands)-coated cell culture plastic and passaged at 70-90% confluency using 0.5 mM EDTA in DPBS (w/o $CaCl_2$, $MgCl_2$; Thermo Fisher Scientific, Waltham, MA, USA) for dissociation, followed by incubation in StemMACS™ iPS-Brew

medium with 1% RevitaCell™ Supplement (#A2644501; Thermo Fisher Scientific) for 18-24 h. Expanded PPMI hiPSCs (1×10^6 cells/vial) were frozen in Bambanker freezing medium (Nippon Genetics Europe, Düren, Germany) and stored at −150°C. Pluripotency of the expanded PPMI hiPSCs (quality control (QC) No. 1) was confirmed by quantitative RT-PCR (RT-qPCR) for pluripotency marker genes (**Table 1**).
The study was conducted according to the guidelines of the Declaration of Helsinki and approved by the Ethics Committee of the Ärztekammer Westfalen-Lippe and Westfälische Wilhelms Universität Münster (protocol code 2017-691-f-S on 2018/01/05).

2. *Directed Differentiation of hiPSCs into mDA Neurons*

Directed differentiation of the PPMI hiPSCs into mDA neurons was conducted in a monolayer culture using a modified protocol from [3–5], as follows. The hiPSCs were dissociated in 0.5 mM EDTA/DPBS (w/o Ca, Mg) and plated onto Matrigel (#356234; Corning) and 0.5 µg/cm^2 human recombinant laminin 111 (#LN111; Bio-Lamina, Sundbyberg, Sweden)-coated tissue culture plastic at a density of 0.5–1.5 × 10^4 cells/cm^2 in N2 medium (1:1 mixture of DMEM:F12 (#21331020; Thermo Fisher Scientific) and Neurobasal (#21103049; Thermo Fisher Scientific), 1% N-2 supplement (#17502001; Thermo Fisher Scientific), 1% GlutaMAX (#35050038; Thermo Fisher Scientific) and 0.2% penicillin/streptomycin (#15140-122; Thermo Fisher Scientific)) with 1% RevitaCell™ for 24 h. Cells were kept in N2 medium for the next 12 days of differentiation (d12) and 100 ng/mL of human Noggin (#130-103-456; Miltenyi Biotec) and 10 µM of SB431542 (#130-106-543; Miltenyi Biotec) BMP/TGF inhibitors were added from d1 to d7 (SB431542) or d9 (Noggin) to this medium. Cells were treated with 375 ng/mL of active human SHH (C24II) (#130-095-727; Miltenyi Biotec) from d3 to d12, 100 ng/mL of human FGF8b (#130-095-740; Miltenyi Biotec) from d3 to d16, and 0.6 µM CHIR99021 (#130-106-539; Miltenyi Biotec) from d2 to d16 of this differentiation protocol. Cells were split using 0.5 mM EDTA/DPBS (w/o Ca, Mg) on d12 and cryopreserved at −150°C in freezing medium (1:1 mixture of DMEM:F12 and Neurobasal, 1% N-2 supplement, 4% B-27 supplement without vitamin A (#12587001; Thermo Fisher Scientific), 1% GlutaMAX and 10% DMSO) on d16 of their differentiation, using StemProAcutase (#A1110501; Thermo Fisher Scientific) for their dissociation. After passaging or thawing, pre-patterned ventral midbrain (VM)/mDA neural stem cells (NSCs)/neural progenitor cells (NPCs) were plated at a density of 2.6 × 10^3 cells/cm^2 in B27 medium (Neurobasal, 2% B-27 supplement without vitamin A, 1% GlutaMAX and 0.2% penicillin/streptomycin) with 1% RevitaCell™ Supplement (for 24 h) and supplemented with 20 ng/mL of human BDNF (#130-096-286; Miltenyi Biotec) and 0.2 µM L-ascorbic acid (#A4403; Sigma-Aldrich, Taufkirchen, Germany) from d12 and with 10 ng/mL of human GDNF (#212-GD-010; R&D Systems, Minneapolis, MN, USA), 500 µM dibutyryl-cyclic AMP (db-cAMP) (#D0627; Sigma-Aldrich) and 1 µM N-[(3,5-difluorophenyl)acetyl]-l-alanyl-2-phenyl]glycine-1, 1-dimethylethyl ester (DAPT) (#2634; Bio-Techne,Wiesbaden-Nordenstadt, Germany) from d16 for the remaining culture time. At d30 (QC No. 2) and d60 (QC No. 3) of this modified differentiation protocol, the pre-patterned VM/mDA

NSCs/NPCs and differentiating mDA precursors and neurons were routinely tested by RT-qPCR for the expression of stage-, brain region- or neuronal cell-type-specific marker genes (**Table 1**).

3. *Reverse Transcription and Quantitative Polymerase Chain Reaction (RT-qPCR)*

Cells at the corresponding stages in 6-well plates were harvested by treatment with 0.5 mM EDTA/DPBS (w/o Ca, Mg; only hiPSCs) or Accutase for ~5 min and pelleting at 350× g for 4 min. Total RNA was isolated with the Monarch Total RNA Miniprep Kit (#T2010S; New England Biolabs) and 1 µg of total RNA was reverse transcribed using LunaScript® RT SuperMix Kit (#M3010L; New England Biolabs), following the manufacturer's instructions. PCR reactions were set up in duplicates in 96-well plates or 0.2 mL tubes using 1 µL of cDNA (5 ng/µL), 1 µM of the primer pairs listed in **Table 1** or **Table 2** (biomers.net GmbH, Ulm, Germany) and PowerUp™ SYBR™ Green Master Mix (#A25742; Thermo Fisher Scientific). Amplifications were performed on a StepOnePlus™ Real-Time PCR System (Thermo Fisher Scientific) with the following parameters: 30s at 95°C, 1 min at 60°C for 40 cycles.

4. *Data analysis*

Each RT-qPCR reaction was performed in duplicate (UNITS in the data sheets). Average Ct values from the duplicates for the corresponding gene, cell type, differentiation experiment and PPMI cell line are included as TESTVALUE in the data sheets. Data labeled as TESTNAME in the data sheets are from three technical replicates (independent differentiation experiments, [6]) for each cell type used for RNA isolation, cDNA synthesis and qPCR (ipsc: undifferentiated human iPSCs; diff1-3 day30: 3 different differentiation experiments on day 30 of the differentiation procedure; diff1-3 day60: 3 different differentiation experiments on day 60 of the differentiation procedure), and human gene detected.

Table 1. Marker gene-specific primer and amplicon sizes for RT-qPCR.

Nr.	Marker Gene (alphabetical)	Forward Primer (5′–3′) Reverse Primer (5′–3′)	Tm (°C)	Product Size (bp)
1	*ACTB (housekeeping)*	GCACAGAGCCTCGCCTT CCTTGCACATGCCGGAG	59.68 58.38	112
2	*ALDH1A1 (mDA)*	ATCAAAGAAGCTGCCGGGAA GCATTGTCCAAGTCGGCATC	59.96 59.90	101
3	*BARHL1 (forebrain)*	GTACCAGAACCGCAGGACTAAA AGAAATAAGGCGACGGGAACAT	60.03 60.09	113
4	*CALB1 (mDA)*	ACGCTGAGCTTTTGCTCACT GCAGGTGGGATTCTGCCATT	60.53 60.69	101
5	*CCK (mDA)*	AAGTGACCGGGACTACATGG TGGGTTGGGAGGTTGCTTC	59.10 59.85	119

6	*DCX (precursors)*	TGCCTCAGGGAGTGCGTTA GAACAGACATAGCTTTCCCCTTC	60.91 59.06	94
7	*DDC (mDA)*	GACACCATGAACGCAAGTGAAT GACCTGGCGTCCCTCAATG	59.77 60.45	93
8	*DRD2 (mDA)*	GCGAGCATCCTGAACTTGTG CTTGGAGCTGTAGCGCGTAT	59.55 60.25	90
9	*EN1 (VM/mDA)*	CGTGGCTTACTCCCCATTTA TCTCGCTGTCTCTCCCTCTC	57.30 60.11	117
10	*ETV5 (VM)*	TCATCCTACATGAGAGGGGGTT GACTTTGCCTTCCAGTCTCTCA	60.02 59.96	120
11	*FOXA2 (VM/mDA)*	CCGTTCTCCATCAACAACCT GGGGTAGTGCATCACCTGTT	57.81 59.67	114
12	*FOXG1 (forebrain)*	TGGCCCATGTCGCCCTTCCT GCCGACGTGGTGCCGTTGTA	66.25 65.70	77
13	*GAD2 (GABAergic)*	CAGGCGACCTGCTCCAGTCTC CGGAATCCCCAGAGCCATCTTC	64.76 62.77	90
14	*HOXA2 (hindbrain)*	CGTCGCTCGCTGAGTGCCTG TGTCGAGTGTGAAAGCGTCGAGG	66.07 65.02	92
15	*LMX1A (VM/mDA)*	ACTCAACAGAGGCGAGCATT TCTGGGTGTTCTGCTGATCT	51.00 49.00	196
16	*LMX1B (VM/mDA)*	CTTAACCAGCCTCAGCGACT TCAGGAGGCGAAGTAGGAAC	59.75 58.82	118
17	*NANOG (pluripotency)*	TGAACCTCAGCTACAAACAG TGGTGGTAGGAAGAGTAAAG	55.32 53.69	138
18	*NES (NSCs/NPCs)*	GCGTTGGAACAGAGGTTGGA TGGGAGCAAAGATCCAAGAC	60.53 57.50	327
19	*NR4A2/NURR1 (mDA)*	GAGACGCGGAGAACTCCTAA CAGGCGTTTTCGAGGAAAT	56.23 58.91	130
20	*OTX1 (VM/mDA)*	TATAAGGACCAAGCCTCATGGC TTCTCCTCTTTCATTCCTGGGC	59.89 60.03	61
21	*OTX2 (VM/mDA)*	ACAAGTGGCCAATTCACTCC GAGGTGGACAAGGGATCTGA	58.38 58.43	62
22	*PAX6 (forebrain)*	TAAGGATGTTGAACGGGCAG TGGTATTCTCTCCCCCTCCT	58.67 57.89	126
23	*PITX3 (mDA)*	AGAGGACGGTTCGCTGAAAAA TTCCTCTGGAAGGTCGCCTC	60.20 61.26	93
24	*POU5F1/OCT4 (pluripotency)*	AGCGAACCAGTATCGAGAAC TTACAGAACCACACTCGGAC	57.44 57.19	118
25	*SLC6A3/DAT (mDA)*	CAAATGCTCCGTGGGACTCA CACTCCGTTCTGCTCCTTGA	60.32 59.68	109
26	*SLC17A6/VGLUT2 (Glutamatergic)*	TGGCAAGAACTGACAACAGTC TCTGGCCGAGTGATTTTCCA	58.71 59.31	109

27	*SLC18A2/VMAT2 (mDA)*	ATGGTCACCGGGAATGCTAC TCTTCACTGGGACAGTCGGA	59.82 60.18	104
28	*SLC32A1/VGAT (GABAergic)*	CACGTCCGTGTCCAACAAGT AGTCGAGGTCGTCGCAATG	60.81 60.15	116
29	*SOX2 (pluripotency, NSCs/NPCs)*	ATGGGTTCGGTGGTCAAGTC GTGGATGGGATTGGTGTTCTC	59.96 58.63	426
30	*SOX6 (mDA)*	CCTTCCCCGACATGCATAACT GTGGATCTTGCTTAGCCGGG	60.13 60.82	119
31	*TH (mDA)*	GGACCTCCACACTGAGCCAT ATGATGGCCTCTGCCTGCTT	61.56 61.93	107
32	*TUBB3 (neuron)*	TTCTCACAAGTACGTGCCTCG GAAGAGATGTCCAAAGGCCCC	60.34 60.68	91

Table 2. Ca^{2+}-associated candidate gene-specific primer and amplicon sizes for RT-qPCR.

Nr.	Candidate Gene (alphabetical)	Forward Primer (5′–3′) Reverse Primer (5′–3′)	Tm (°C)	Product Size (bp)
1	*ANO4*	AATGGAGGCAAGCTCTTCTGG ACATCACTCTTTGGCCCGTC	60.34 60.32	126
2	*CACNA1A*	AAGGAGAGGAGGATGCGTTTC CAGCGTGTTGAGAGCTACCAAA	59.79 61.12	93
3	*CACNA1B*	CATCAACCGCCACAACAACT ACATCAGAAAGGAGCACAGG	59.33 57.51	206
4	*CACNA1C*	GCAATCACCGAGGTAAACCCA CAGTTAGCACAGCAAGAGGG	60.61 58.55	91
5	*CACNA1D*	AGCCATCTCAAAATCCAAACTCA CACGGCGGCCCTACATCT	58.59 61.85	79
6	*CACNA1E*	TCCAGTTGGCTTGTATGGAC AACGTCTCATGGAGCTAGGG	57.51 58.88	90
7	*CACNA1H*	TCAACGTCATCACCATGTCC GCCTCGAAGACAAACACGA	57.91 58.10	106
8	*CACNA2D1*	CTGACGGTCCAAATCCTTGT GTCATAACAGGCGGTGTGTG	57.81 59.20	201
9	*CACNA2D3*	GAACATCCCGATGTGTCCTTGG ACTGGAGCAGAGGTTCTTTGCC	61.26 62.78	130
10	*CACNB3*	GTACCAGCCTCACCGCCAAC CCGGTCACTGTGGTTGTGCT	63.04 62.94	106
11	*CACNG2*	CAAAACTCGACACAACATCATCC GGACCAGCCGTATGAGTAACT	58.53 59.24	154
12	*CACNG3*	CTCCCGGGACCCCTCAAAGA GAGGTCAGTTCAGACGGGCG	63.11 62.82	157

13	*CACNG4*	CACTTCCCAGAGGACAATGACTAC GTAAAAAGACCAGCCGTAGTTGTAA	60.62 59.53	306
14	*CHRM5*	AAGTACGCTCACCGTCCAGA AGGCATTTCTTCTGTCCTCCTC	60.89 59.76	126
15	*CHRNA3*	TCCGGATGGTTCCTGTCCTC CAGACCTGGAGCCGTGC	60.98 60.09	105
16	*CHRNA6*	TATTCGCCCTTCATGTCTGCC ACTGCAATCCGTGTGTGTCT	60.48 59.89	102
17	*CHRNB4*	TCCTTTTCTTCCTGGTCGCC GGAGATGAGCTGTGAGGAGC	59.96 59.90	143
18	*CNR1*	CCCTGTGGGTCACTTTCTCAG GATGGTGCGGAAGGTGGTAT	60.27 59.82	104
19	*CNRIP1*	AGACACGCAGTCTGATGTGG GGCCTGCGCTGGTTTATCTA	55.00 55.00	113
20	*FXYD5*	AGACCCTCAAGCCATCTGGT ACTTGCCACTGGTGAGGATG	60.55 59.96	132
21	*FXYD6*	TTGGGACCTGGGAAGGTTTG TCAGACTGGGATCAGGTGGG	59.81 60.62	105
22	*FXYD7*	AATGACGATCCCCCAAAGAGC TTCGGGGATGAGACAAGGAG	60.41 58.80	145
23	*GABBR2*	CCTCATGCCGCTCACCAA GAGTGACTCGTTGCGGATCT	60.05 59.83	100
24	*GABRA2*	GGCTTGGGATGGGAAGAGTG GGGCATAATTGGCAACAGCC	60.39 60.18	101
25	*GABRA3*	GGTCTCTCCAAGTTTGTGCCT CCCGGGTTCTTGTCGTCTTG	60.20 60.95	141
26	*GABRA5*	CGATCCTCCAGCCCAGAGAC CTGAAATATCTGCCACGCCAC	61.74 59.67	102
27	*GABRB2*	AAAGGGATGTGGAGAGTGCG ACCGTCTCTTTAACCAGCGA	60.04 59.04	125
28	*GABRQ*	CCTCGCTGCTCTCTCCTTAGA GCCGCGAAAAGGAACAGAG	60.75 59.50	112
29	*GLRA2*	ACCCCCTTCAGTATCCCCTC CAGTTGCTTGGGGCGGG	60.03 61.08	106
30	*GRIA2*	GGGCGCCGATCAAGAATACA AGCTGTTTGCCACCTCCAAA	58.67 59.97	107
31	*GRIA3*	TTGACGACTCCTGAGTTGCG TTGGGGAATCCTCCGTGAGA	60.32 60.25	149
32	*GRIA4*	TGAAACTCAGTGAGGCAGGC TCAAGGCACTCGTCTTGTCC	60.25 59.97	111
33	*GRIK1*	AAGCAAAGATTCAGAGAAGCCG AGGGTACAGGGTGAGTTGGT	59.25 60.10	103

34	*GRIK3*	AGCAAAGCAGGAGACCACAG TCCTCTCCCTCGTCCAATGT	60.25 59.96	137
35	*GRIN1*	CGACCCCAAGATCGTCAACA TGAATCTTCCAGGAGCCGTG	60.04 59.75	111
36	*GRIN3A*	AAGGCAGAGAGGAAGTAGCG GGTGCGGAAAAGAAGCCAAAT	59.18 60.00	111
37	*GRIN2B*	CGGCTTCTTCACCTCTAGGAAA GATATGCATTCGGACGCCAG	59.77 59.21	109
38	*GRM2*	AGCAGGAGTCCAAGATCATGTTT AAAGTCCTTGTAGAGGCGGC	59.99 60.04	149
39	*GRM7*	ACATCAAGAGGGAAAACGGGA GTTCGAGCGCGTAAGTGTCC	59.30 61.67	147
40	*HCN1*	CTTTCCACTATGCCTCCCCC CATTTTTCGGCGTGGAGCTG	59.82 60.45	150
41	*KCNB2*	ACGAAGTCCTGTGGAGAACG ATAGTCGTCGCACACTTCCA	59.69 59.11	104
42	*KCND2*	TTGCTCTACCTGTTCCGGTG TGTAAGCATTTGCGCTTCCG	59.68 59.83	141
43	*KCND3*	TGTACGAACCTCCACCATCAA ATGGGTAGTTCTGCATTGAACTCT	59.03 60.02	92
44	*KCNG3*	ACGAGATGATCTACTGGGGC GCCGAGTAGAAGGTGTAGGTG	58.67 59.87	100
45	*KCNH7*	CTCTAGCCGAGGAGAGATCC TTTCCGAATGATGGTCCCCA	58.18 59.00	135
46	*KCNIP1*	CAGGTCCTTTATCGAGGCTTCA CGACAGAGCGGTTACAAAGTC	59.83 59.28	189
47	*KCNJ3*	GTGGAAACAACTGGGATGAC GTTGCATGGAACTGGGAGTA	56.63 57.51	137
48	*KCNJ6*	TCCAGCAAACTCAACCAACATGC GCCCAGCTAGGGCACTAAAC	62.35 60.75	131
49	*KCNK9*	GCTCCTTCTACTTTGCGATCACG CTGGAACATGACCAGTGTCAGC	61.82 61.45	134
50	*KCNMA1*	TCACGCAGCCCTTTGC CCGTATCAGGGTGAGGATATTG	57.72 57.81	101
51	*KCNMB2*	TGAATGCGTCCATCACGGAA TCCCCGGAAGAAGTCAGGTT	50.00 55.00	115
52	*KCNMB4*	GATTGGTTCCCAGCCATTTAC AATGAGAACGCCCACCAC	57.20 57.60	142
53	*KCNN2*	TGGTAGCTGTAGTGGCAAGGA TGTTTCCCTGAGTACATTGGC	60.83 58.21	125
54	*KCNQ2*	AGCATCCTCAGCAAACCTCG TAGGCGTGGTAGATGAACGC	60.39 59.90	140

55	*KCTD6*	GTCTCTTAGTTCCAGATGGGTTCTC GCTGAAACCTACTACAGGGCA	60.39 59.72	106
56	*NCS1*	CCCACCAAGTTTGCCACATTT CATTGTCCAAGTCGTAGAGCTT	59.86 58.41	157
57	*SCN1A*	ACCATAGAGTGAGGCGAGGA TACAAGGGCTGCAGTCTCAC	59.74 59.68	128
58	*SCN2A*	TCATGGCATCAAACCCCTCC GGTAGCGTCTGTAAGCCCTC	60.03 59.90	111
59	*SCN3A*	TCCGGAAAAAGGTGCTGTGA AAAGGCGGAAGCTTTCAGGT	59.82 60.18	128
60	*SCN3B*	GACCGACGTGGAACGCAT CTGCTTCGTCCGCCCTAAC	60.43 60.81	101
61	*SCN9A*	CGAAGGCAGTGCAGTCACTA TCCTACAGAGTTCTCTTACGATTCT	60.04 58.58	111
62	*TRPC4*	CTCGGAAGAACTGCTCTCCTC CTCAACAGCTCCGACGACTT	59.87 60.04	135
63	*TRPC5*	GCAAAAGAGGAGTCCGGCAA GCAATGCAGAAATCCTGAGCC	60.89 60.20	109
64	*TRPV2*	TGTCCTGACAGGGGAGAGTTA TAAGCTCCAGCAAGCAGGTT	59.57 59.60	100

References

1. Nouri, P.; Götz, S.; Rauser, B.; Irmler, M.; Peng, C.; Trümbach, D.; Kempny, C.; Lechermeier, C.G.; Bryniok, A.; Dlugos, A.; et al. Dose-Dependent and Subset-Specific Regulation of Midbrain Dopaminergic Neuron Differentiation by LEF1-Mediated WNT1/b-Catenin Signaling. *Front Cell Dev Biol* **2020**, *8*, 587778, doi:10.3389/fcell.2020.587778.
2. Nouri, P.; Zimmer, A.; Brüggemann, S.; Friedrich, R.; Kühn, R.; Prakash, N. Generation of a NES-mScarlet Red Fluorescent Reporter Human iPSC Line for Live Cell Imaging and Flow Cytometric Analysis and Sorting Using CRISPR-Cas9-Mediated Gene Editing. *Cells* **2022**, *11*, 268, doi:10.3390/cells11020268.
3. Nolbrant, S.; Heuer, A.; Parmar, M.; Kirkeby, A. Generation of high-purity human ventral midbrain dopaminergic progenitors for in vitro maturation and intracerebral transplantation. *Nat Protoc* **2017**, *12*, 1962–1979, doi:10.1038/nprot.2017.078.
4. Kriks, S.; Shim, J.W.; Piao, J.; Ganat, Y.M.; Wakeman, D.R.; Xie, Z.; Carrillo-Reid, L.; Auyeung, G.; Antonacci, C.; Buch, A.; et al. Dopamine neurons derived from human ES cells efficiently engraft in animal models of Parkinson's disease. *Nature* **2011**, *480*, 547–551, doi:10.1038/nature10648.
5. Rus Jacquet, A. de. Preparation and Co-Culture of iPSC-Derived Dopaminergic Neurons and Astrocytes. *Curr Protoc Cell Biol* **2019**, *85*, e98, doi:10.1002/cpcb.98.
6. Anderson, N.C.; Chen, P.-F.; Meganathan, K.; Afshar Saber, W.; Petersen, A.J.; Bhattacharyya, A.; Kroll, K.L.; Sahin, M. Balancing serendipity and reproducibility:

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